Supplementary Material for CompositionEvidence: Controlling Geometric Evidence for Cinematic Composition Reasoning with Frozen Vision–Language Models

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Anonymised for double-blind review

Abstract. This document reports before-and-after question-answer examples, a counterexample, the exact controlled-text interface, feature and calibration definitions, soft-token training details, additional per-class and corruption results, the artifact map, and a reproducibility summary. The main paper contains the primary method and results; every number here is recomputed from the per-sample records in the accompanying anonymous code package.

1 Contents and Notation

Sections 2–3 report concrete Q&A transitions, including one failure case. Sections 4–6 specify the controlled-text and soft-token interfaces. Sections 7–8 provide additional results and implementation details.

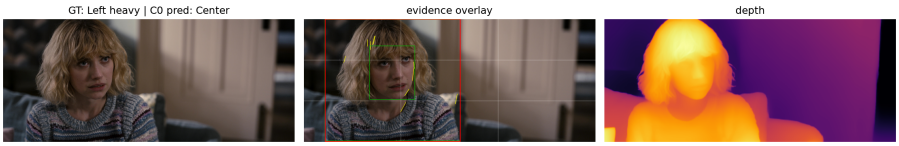
Notation. "Before" denotes the official zero-shot prompt (C0). "After-A" denotes the training-free, post-question, confidence-gated text channel. "After-B" denotes four learned soft tokens with all VLM weights frozen. The red box in each overlay is the selected dominant subject; green boxes are other detections, white lines are the thirds grid, yellow segments are detected lines, and the third panel is relative depth (brighter means closer).

2 Before-and-After Q&A Demos

The following examples were selected from a fixed pool of C0-wrong/After-A-correct items. Their questions, options, predictions, and measurements are

Table 1. Primary Qwen2.5-VL-7B results reproduced from per-sample logs.

Auditable claim	Correct	Acc.
Zero-shot C0	237/479	49.5
Naive pre-question text	222/479	46.3
Channel A: post-question + gates	269/479	56.2
Channel B: four soft tokens	292/479	61.0



Question. What's the composition of this shot? **A** Left heavy **B** Balanced **C** Symmetrical **D** Center

Ground truth	**A — Left heavy**
Before (C0)	**D — Center**. "The subject is positioned in the center of the frame."
Measured evidence	Dominant-person centroid x=0.31; subject area 46%; left/right visual mass 0.79/0.21; absolute difference 0.59 (96th train percentile).
After-A / After-B	**A — Left heavy** / **A — Left heavy**
What changed	The baseline focused on the face and ignored the full subject footprint. The evidence exposes a large frame-level mass displacement toward the left.

Fig. 1. Demo 1: local face-centering versus frame-level mass. The full-body subject box and mass estimate correct a center-biased visual prior. Sample `shotbench-2631`.

copied from the dataset and run logs. They illustrate three different failure modes rather than three near-duplicate images.

What each demo isolates. Demo 1 separates local face-centering from the visual mass of the full subject. Demo 2 separates negative space from occupied visual weight and therefore tests the sign of the geometric correction. Demo 3 separates semantic grouping of two people from the geometry of the selected foreground subject; it also records a disagreement between the two proposed interfaces.

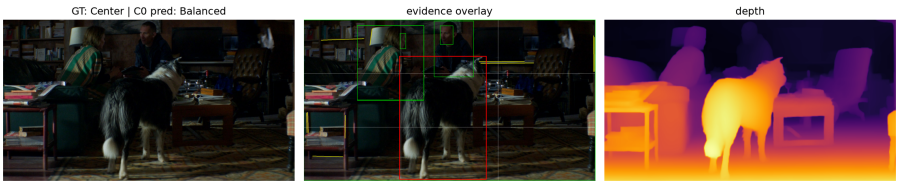
How to interpret the examples. The overlays are diagnostic visualisations and are not given to the VLM. The model receives the unchanged film still plus either controlled text (Channel A) or soft tokens (Channel B). Each "What changed" statement is a post-hoc reading of the logged measurements and predictions. The performance claim rests on the aggregate paired tests in Section 7; Section 3 gives a case in which Channel A turns a correct answer into a wrong one.



Question. What's the composition of this shot? **A** Right heavy **B** Symmetrical **C** Left heavy **D** Short side

Ground truth	**A — Right heavy**
Before (C0)	**C — Left heavy**. "The person is positioned on the left side of the frame, and there is more space on the right side."
Measured evidence	Dominant-person centroid x=0.64; subject area 42%; left/right visual mass 0.36/0.64; difference 0.27 (78th train percentile); right edge distance 0.14.
After-A / After-B	**A — Right heavy** / **A — Right heavy**
What changed	The baseline conflated negative space with visual weight. The measured centroid and mass distinguish empty space from the occupied, high-saliency half.

Fig. 2. Demo 2: negative space versus visual weight. Both calibrated interfaces reverse the baseline's directional error. Sample `shotbench-2638`.



Question. What's the composition of this shot? **A** Balanced **B** Short side **C** Center **D** Symmetrical

Ground truth	**C — Center**
Before (C0)	**A — Balanced**. "The person on the left and the person on the right [create] a balanced composition."
Measured evidence	The selected foreground animal has centroid $(0.48, 0.61)$ and occupies 23% of the frame; left/right mass is near-neutral at 0.53/0.47; left–right mirror SSIM is low (0.21).
After-A / After-B	**C — Center** / **A — Balanced**
What changed	Channel A makes dominant-subject selection explicit and corrects the two-person grouping heuristic. Channel B does not correct this item, showing that the two interfaces learn different trust behaviours rather than identical rules.

Plain-language decision trace.

System	What it prioritises	Decision shortcut	Output
Before (C0)	Two visible people	"Two people on opposite sides" ⇒ balanced	**A — Balanced**
Channel A	Explicit dominant-subject geometry	Foreground animal is centred; weak mirror evidence is not rendered	**C — Center**
Channel B	The same 24 measurements, encoded as four learned tokens	Soft evidence does not outweigh the baseline semantic grouping on this item	**A — Balanced**

Interpretation. The two interfaces share an evidence source but not a decision rule. The disagreement is therefore diagnostic. A correct Channel-A transition does not imply that every learned integration route makes the same transition.

Fig. 3. Demo 3: semantic grouping versus dominant-subject geometry. This is also a disagreement case: Channel A succeeds while Channel B retains the baseline answer. Sample `shotbench-2692`.

3 Counterexample: Multi-Person Composition

A selected failure case accompanies the three corrected examples. For sample `shotbench-2632` the options are *A Symmetrical*, *B Left heavy*, *C Balanced*, and *D Short side*, with ground truth C. C0 answers C correctly; After-A changes the answer to B, while After-B remains correct at C. The extractor selects one face at $(0.45, 0.45)$ in a multi-person scene and reports right-skewed mass (0.36/0.64). Reducing the scene to one dominant face removes the relational layout that made the original answer correct.

This case motivates two limitations. First, dominant-subject synthesis should become set-valued when several similarly salient actors form the composition. Second, the controlled-text channel exposes measurements but cannot learn all interactions among them; the soft-token projector can sometimes suppress a misleading feature combination, but Section 7.3 shows that it has different weaknesses.

4 Exact Channel-A Interface

4.1 Prompt order

The official C0 order is

$$I \rightarrow q \rightarrow \text{options} \rightarrow \text{answer instruction}.$$

Naive injection places the evidence block before q. Channel A preserves the official sequence and appends the block:

$$I \rightarrow q \rightarrow \text{options} \rightarrow \text{answer instruction} \rightarrow \text{evidence}.$$

No test label, predicted label, or answer-specific template is used. The final line states that the auxiliary measurements may be wrong and requests an option letter.

4.2 Exact rendered block for Demo 2

The following is the final gated block appended for `shotbench-2638`. The mirror-similarity line is suppressed because its horizontal SSIM lies inside the uninformative interval.

Table 2. An intentionally reported counterexample. A single-subject abstraction is not always sufficient for multi-person composition.

Condition	Prediction	Status
C0	C — Balanced	**correct**
After-A	B — Left heavy	**wrong**
After-B	C — Balanced	**correct**

```
<composition_evidence>
Subject: one dominant subject (person), centroid at
(0.64, 0.52), occupying 42% of the frame; 0.14 from
the frame midpoint, 0.15 from the nearest thirds
intersection.
Edges: distance from subject box to frame edges -
left 0.42, right 0.14, top 0.05, bottom 0.00
(subject spans the full frame height).
Visual mass: left/right = 0.36/0.64 (difference 0.27
(78th percentile of film frames), mass concentrated
in the right half).
Depth: subject in mid-ground; foreground 19% / mid
28% / background 54% of pixels.
Lines: 10 of the 10 longest line segments pass
through the subject region when extended, dominant
angle -85 deg.
Head space: top of face at y=0.13 (headroom 13% of
frame height).
</composition_evidence>
Note: the evidence above is computed by auxiliary
vision tools and may contain errors; use it as
reference alongside the image. Answer with the
option letter only.
```

4.3 Leakage guard

The renderer hard-fails if an evidence line contains a root shared with the benchmark's answer vocabulary. For ShotBench the guarded roots cover *center/centre/central*, *symmetr-*, *balanc-*, *heavy*, *short side*, *rule of thirds*, and *golden ratio*. The guard is derived again for a benchmark with a different label taxonomy. Percentiles are computed only from the ShotQA training distribution.

5 Geometry, Gating, and Calibration

5.1 Feature definitions

5.2 Dominant-subject and pseudo-saliency synthesis

GroundingDINO is queried for people/faces and then with a generic-subject fallback. Boxes covering more than 90% of the frame are discarded as non-positional. The dominant subject maximises a fixed combination of box area, centre prior, and detection confidence with weights 0.4/0.3/0.3; person terms receive a 1.2× multiplier. Pseudo-saliency averages a confidence-weighted detection mask and normalised depth closeness with equal weights. Depth Anything V2 and OpenCV LSD supply the remaining frozen observations.

5.3 Training-set calibration

Calibration uses 1,328 ShotQA training images (seed 42, at most 300 per class). No ShotBench test label enters the numeric fit. Youden-J fitting gives the final mass threshold $\tau_{\text{mass}} = 0.143$ and upper mirror threshold $\tau_{\text{mirror,hi}} = 0.262$; the lower mirror gate is the training-distribution 10th percentile, $\tau_{\text{mirror,lo}} = 0.144$.

Table 3. The nine formalised features and the final Channel-A gates, expanding the summary table in the main paper. Channel B always receives the corresponding continuous values plus subject/depth presence flags.

ID	**Quantity**	Computation and Channel-A rule
F1	**Subject centroid**	Normalised box centre $(x, y) \in [0, 1]^2$; emit when a subject exists.
F2	**Reference distances**	Distance to midpoint, nearest thirds intersection, and thirds line; emit with F1 without answer words.
F3	**Subject area**	Box area divided by frame area; emit with F1.
F4	**Mirror similarity**	Left–right and top–bottom SSIM after fixed resizing; emit only when horizontal SSIM is ≤ 0.144 or ≥ 0.262.
F5	**Visual mass**	Confidence-weighted boxes plus depth closeness; report left/right shares and $b \in [-1, 1]$ only when $\lvert b\rvert \geq 0.143$.
F6	**Relative depth**	Equal-width closeness bins and subject median layer; emit when depth exists.
F7	**Lines and horizon**	Ten longest LSD segments; intersection with a 1.2× dilated subject box; emit when extraction succeeds.
F8	**Head space**	Face-top coordinate and top-edge distance; emit when a face is detected.
F9	**Edge proximity**	Four frame-edge distances; emit only within 0.05 of an edge or when spanning a frame dimension.

The edge gate uses 0.05. Pseudo-saliency direction precision on calibration data is 67.8%, which motivates gating rather than unconditional prose.

The choice to add calibration and selective rendering followed first-pass ShotBench diagnostics. Thus, the thresholds are fitted on training data, but the interface design is adaptive to the target benchmark. The reported final runs use the fixed values above.

A stricter precision-oriented variant uses $|b| \geq 0.4$ and horizontal SSIM ≥ 0.65 (or ≤ 0.144), reaching 54.5%. The final Youden-gated variant reaches 56.2%. A 3×3 sweep over the centre/thirds alignment threshold (0.05–0.12) and the strong-symmetry threshold (0.6–0.8) records 56.8–57.0%, so the qualitative conclusion does not depend on one exact threshold pair.

6 Channel-B Architecture and Training

6.1 Feature vector and projector

The 24-dimensional vector is

$$\begin{aligned}\mathbf{z} = [&x, y, d_{\text{mid}}, d_{\text{thirds}}, d_{\text{thirds-line}}, a, e_l, e_r, e_t, e_b,\\ &b, m_l, m_r, s_h, s_v, d_f, d_m, d_b, n_{\text{lead}}/10,\\ &\theta/90, y_{\text{horizon}}, y_{\text{face}}, \nVdash_{\text{subject}}, \nVdash_{\text{depth}}].\end{aligned}$$

Missing continuous values are zero-filled and paired with the applicable presence flag. A GELU MLP maps

$$24 \to 512 \to 4d,$$

where $d = 3{,}584$ is the Qwen2.5-VL-7B hidden size. The output is reshaped into four token embeddings. A forward hook replaces four rare placeholder-token embeddings before the question; visual and official prompt tokens are otherwise unchanged.

Table 4. Controlled variants on all 479 ShotBench composition items.

Condition	Correct	Acc.
C0 zero-shot	237	49.5
Naive text, pre-question	222	46.3
Text, post-question	256	53.4
Post-question + strict gates	261	54.5
Channel A: Youden gates	**269**	**56.2**
Channel B: four tokens	**292**	**61.0**
Channel B: eight tokens	288	60.1

Parameter count. The implemented projector has

$$(24\times512+512)+(512\times4d+4d)=7{,}367{,}168$$

trainable parameters (7.37M). The VLM, vision encoder, tokenizer, and perception experts remain frozen.

6.2 Training protocol

The source pool contains 9,626 ShotQA composition examples. A seed-42 shuffle makes a 9,145/481 train/validation split. The projector is FP32; its outputs and the frozen VLM are BF16. Optimisation uses AdamW (learning rate 10^{-4}, default weight decay 0.01), batch size 1, gradient accumulation 8, gradient-norm clipping at 1.0, a CosineAnnealingLR scheduler, and two epochs. Training images use a $768\times28\times28$ pixel budget. The loss is language-model cross-entropy on the final answer token(s); validation loss selects the checkpoint.

Evaluation uses greedy decoding, at most 64 new tokens, and the official option format. A one-epoch eight-token variant reaches 60.1%, compared with 61.0% for the final four-token setting.

7 Additional Results

7.1 Position, gating, and learned integration

Against C0, Channel A changes 100 decisions: 66 wrong-to-right and 34 right-to-wrong (exact McNemar $p = 0.0018$). Channel B changes 145 decisions: 100 wrong-to-right and 45 right-to-wrong ($p < 10^{-5}$). Naive pre-question injection changes 165 decisions (75 wrong-to-right, 90 right-to-wrong; $p = 0.276$). The transition counts distinguish paired improvement from a change in aggregate percentage alone.

7.2 Corruption controls

Corruption randomises the subject centroid, re-derives dependent distances, flips the visual-mass direction, and randomises mirror similarity. The naive interface

Table 5. Internally consistent corruption controls. The four-option chance rate is 25%; 33.6% is above chance but 15.9 points below C0.

Evidence condition	Correct	Acc.
C0, no evidence	237	49.5
Naive, clean	222	46.3
Naive, all fields corrupted	161	33.6
Channel A, clean	269	56.2
Channel A, position corrupted	259	54.1
Channel A, mass corrupted	245	51.1
Channel A, mirror corrupted	270	56.4
Channel A, all corrupted	232	48.4

Table 6. Accuracy (%) for the six non-composite labels. Composite-label items ($n = 21$) are omitted because individual combinations have only one to eight samples.

Ground-truth class	n	C0	Channel A	Channel B
Balanced	73	74.0	57.5	**90.4**
Center	83	65.1	59.0	**73.5**
Left heavy	81	38.3	**75.3**	71.6
Right heavy	79	48.1	73.4	**79.7**
Short side	69	0.0	0.0	1.4
Symmetrical	73	**65.8**	64.4	34.2

therefore follows wrong evidence strongly. The final interface falls back close to the no-evidence baseline when all fields are corrupted. Mass corruption has the largest single effect, consistent with the remaining class-specific errors.

7.3 Per-class behaviour

Channel A's gains come mainly from directional composition, while it loses 16.5 points on *Balanced*. Channel B recovers *Balanced* strongly but loses 31.6 points on *Symmetrical*. Both channels remain nearly unable to select *Short side*, which is a terminology and coverage limitation.

7.4 Model and benchmark heterogeneity

The nine-model comparison appears in the main paper. Whether the same Channel-A prompt improves a model depends on that model's evidence-integration behaviour, and the five commercial models were queried through official provider APIs. On the 120-item CineTechBench composition subset, C0 is 65.0% and Channel A is 64.2% (one item lower). The leakage guard also removes useful thirds-related wording when that phrase appears in the target label taxonomy.

8 Reproducibility and Artifact Map

8.1 Runtime and deterministic settings

Local runs use Python 3.12.3, PyTorch 2.12.1 with CUDA 13.0, Transformers 4.57.1, and one NVIDIA RTX 4090 D (24GB). Inference uses greedy decoding and seed 42. All 479 composition samples are evaluated once per deterministic local condition. Commercial-model identifiers, official provider APIs, and evaluation dates are retained in the logs. Exact McNemar tests use per-sample paired correctness.

8.2 Artifact map

`configs/default.yaml`: frozen thresholds, extractor settings, seed, and decoding.
`configs/calibration.json`: train-only percentiles, thresholds, and gate metadata.
`configs/prompts.yaml`: official prompt, evidence templates, and banned roots.
`results/evidence/shotbench_sc.jsonl`: per-sample features and rendered measurements.
`results/eval/*_C0_sc.jsonl`: zero-shot responses and correctness.
`results/ablation/*_gatedJ_after_sc.jsonl`: final Channel-A predictions.
`results/eval/*_C4_sc.jsonl`: final Channel-B predictions.
`results/projector/best.pt`: validation-selected projector weights.
`scripts/run_final.py`: Channel-A evaluation entry point.
`scripts/train_projector.py`, `scripts/eval_c4.py`: Channel-B training and evaluation.
`tests/`: synthetic feature and renderer-guard tests.

8.3 Double-blind and leakage checklist

- This document and the artifact paths contain no author names, institutions, acknowledgements, personal URLs, or machine-specific absolute paths.
- Gates, percentiles, and thresholds are fitted from ShotQA training data; no ShotBench test label enters that fit or feature construction. The interface design followed initial ShotBench diagnostics, as disclosed above.
- The image, official prompt, and answer options are unchanged across C0 and Channel A; only the post-question evidence block is appended.
- All VLM parameters remain frozen in both channels. Channel B stores only the 7.37M-parameter projector.
- Per-sample JSONL files permit independent reconstruction of every aggregate accuracy, paired transition count, and qualitative example in this supplement.

9 Reproducibility Summary

The following statements summarise what a reviewer can verify from the accompanying anonymous code package without access to any private resource.

Item	Status
Conceptual outline of the method in the main paper	yes
Theoretical contributions requiring proofs	none
Motivation for the selected benchmarks	yes
Benchmarks drawn from published work, cited and public	yes
New benchmark or dataset introduced	none
Pre-processing, evaluation, and analysis code included	yes
Final hyper-parameters listed (Section 6)	yes
Range of values tried per hyper-parameter	partial
Seeds and deterministic decoding described	yes
Computing infrastructure and library versions stated	yes
Number of runs per reported result (one per condition)	yes
Analysis beyond aggregate means (paired transitions, per class)	yes
Significance judged with an appropriate paired test	yes
Code released for research use upon publication	yes

Two limits deserve explicit mention. First, the search over hyper-parameters was not exhaustive: the reported gates come from a Youden-J fit and a nine-point sweep, and we trained the projector at a single learning rate with two token counts. Second, we added the gating mechanism after a first pass of ShotBench diagnostics, so ShotBench is not a fully independent confirmatory evaluation for that design choice. The numeric thresholds use ShotQA training data only.

10 Code Package

The submitted archive `CompositionEvidence_MMM27_Anonymous_Code.zip` contains the extraction, evidence, evaluation, and training code, the validation-selected 7.37M-parameter projector, the per-sample result records for every condition reported in the paper, and the unit tests. Two scripts recompute the headline and reported numbers directly from the stored records without any model download:

```
python scripts/recompute_headline.py  python
       scripts/recompute_reported.py
```

The archive excludes benchmark media, model weights other than the projector, version-control history, and machine-specific paths. Commercial API keys are supplied through environment variables and are not present in the archive.